

Ahmed Khalid Ismail Mohamed

First of Al-Haram, Giza, Egypt • +201127695327 • ahmed.khalid.mohamed7@gmail.com
[linkedin.com/in/ahmedkhalid7/](https://www.linkedin.com/in/ahmedkhalid7/) • github.com/ahmedkhalid7

Objective

Detail-oriented IT professional with a strong foundation in System Administration, DevOps practices, and full-stack development. Experienced in diagnosing and resolving hardware, software, and network issues, with hands-on exposure to Linux/Windows administration, AWS cloud, and containerized environments (Docker/Kubernetes). Proven ability to support and maintain enterprise applications through automation, monitoring, and CI/CD pipelines, with a focus on ensuring high availability, security, and performance. Looking forward to collaborating with forward-thinking teams.

Education

Bachelor of Science in Computer Systems Engineering October University for Modern Sciences and Arts (MSA)

Sep 2019 – July 2024
Giza, Egypt

- Ranked 1st in the class with a GPA of 3.79/4.0 (Excellent with First-Class Honours).
- Graduation Project: Heart Rate Estimation from Facial Videos using Deep Learning Graded "Excellent"; research paper published on the IEEE conference 2024 Intelligent Methods, Systems, and Applications (IMSA) · Sep 6, 2024
- Received dual degree validation from the University of Greenwich in the UK.

Experience

IT Help Desk Egyptian Armed Forces

Dec 2024 – Dec 2025
Beni Suef, Egypt

- Troubleshooting hardware, software, and networking issues for PCs and printers.
- Updated database records, setting up day-to-day work schedules using MS Word and Excel.
- Managed organizational unit archive office for document processing, organization and retrieval.
- Responding to user requests via verbal communication, written documents, or faxes, and escalating issues when necessary.

DevOps Engineer Trainee Digital Egypt Pioneers Initiative

June 2024 – Oct 2024
Cairo, Egypt

- Performed Linux system administration, configuration management, and automation using Shell utilities.
- Streamlined infrastructure configuration using Terraform (IaC) and Ansible roles.
- Achieved a scalable and fault-tolerant system by optimizing Docker containerization and Kubernetes orchestration.
- Optimized GitHub Actions CI/CD pipelines for streamlined service releases on AWS.
- Deployed the Prometheus-Grafana stack for cluster workload monitoring and metrics visualization.

Software Engineer Intern Fawry Banking and Payment Technology Services

July 2023 – Oct 2023
Giza, Egypt

- Gained foundational experience in OOP, SOLID principles, and design patterns through full-stack development training.
- Developed backend services using Java (Spring Boot, JPA, REST APIs) and applied JUnit/Mockito for effective unit testing.
- Built a microservices-based eCommerce platform (API Gateway, Service Discovery, Bank API, User API, etc), integrating secure JWT authentication with an Angular frontend.
- Applied Docker containerization and DevOps best practices, laying the groundwork for continuous integration exposure.

Information Technology Intern Contact Financial Holding

July 2023 – July 2023
Cairo, Egypt

- Engaged with cross-functional technical teams to analyze industry trends and career growth opportunities.
- Collaborated with the DevOps team on best practices for containerization, orchestration, and CI/CD pipelines.

Projects

End-to-End DevOps Implementation for Microservices-based eCommerce Web App

Oct 2025

- Implemented end-to-end DevOps practices for a multi-stack, microservices-based eCommerce web app.
- Ensured consistent infrastructure provisioning with Terraform using remote state management (S3) and state locking (DynamoDB), and developed reusable modules for AWS VPC and EKS.
- Deployed Kubernetes workloads on AWS EKS, configuring Ingress rules to route traffic to backend microservices.
- Implemented CI/CD automation using GitHub Actions and ArgoCD GitOps, resulting in streamlined software releases.
- Configured Prometheus and Grafana stack for k8s cluster monitoring and Loki for log aggregation.
- Configured K8s Gateway API using Istio Service Mesh controller for traffic management and monitoring.

E-Commerce Store Management System using Java Spring Boot, Angular, and MySQL

Nov 2023

- Developed a full-stack eCommerce web application with user and admin interfaces for product browsing, purchasing, and store management.
- Built and integrated RESTful APIs using Spring Boot, and developed frontend with Angular.
- Implemented order creation and checkout logic that integrates with APIs of other microservices.
- Tested and validated APIs using Postman.

Heart Rate Estimation from Facial Videos using Deep Learning (Graduation Project – research paper published on the IEEE conference 2024 Intelligent Methods, Systems, and Applications (IMSA) Sep 6, 2024)

June 2024

- Developed a real-time, contactless heart rate estimation system using deep neural networks to analyze facial color intensity variations linked to blood volume pulse signals.
- Designed a 2D CNN architecture with dual facial regions (cheeks and forehead) for feature extraction, achieving 33 ms average inference time and outperformed state-of-the-art benchmarks in accuracy.
- Trained and deployed the model on Google Cloud Platform (GCP) with an Angular frontend and Flask backend for live web-based inference.
- Performed unit, integration, and stress testing to ensure scalability and reliability in real-time operation.

3-Tier Enterprise LAN Network Implementation

Mar 2025

- Implemented a 3-tier enterprise network (Access, Distribution, Core) supporting scalable architecture.
- Configured VLANs, trunk ports, and network protocols (SSH, NTP, HSRP, VTP, Syslog) for fault tolerance and high availability.

Express.js Application Deployment on Google Kubernetes Engine (GKE)

Apr 2025

- Containerized an Express.js + MongoDB application using Docker for scalable cloud deployment.
- Deployed and managed workloads on GKE with Kustomize for environment-specific configuration.

8-bit Computer using Arduino and Logic Gates

Jan 2023

- Built a Turing-complete 8-bit computer from logic gates and Arduino, implementing custom fetch-decode-execute cycles.
- Designed ALU, RAM, control unit, and bus system demonstrating low-level computing fundamentals.

Smart Home System (Embedded-C, Proteus)

June 2022

- Developed an embedded system using dual ATmega32 microcontrollers communicating via SPI protocol.
- Integrated timers, LCD, and keypad modules to manage home automation functions.

Courses & Certificates

Microsoft Certified Solutions Associate (MCSA) — Eng. Ahmed Nazmy (Dec 2025)

AWS Certified Cloud Practitioner — Udemy (Oct 2025)

Docker and Kubernetes: Zero to Hero — Udemy (May 2025)

GitHub Actions: Beginner to Expert — Udemy (May 2025)

Cisco Certified Network Associate (CCNA) — Jeremy's IT Lab (Feb 2025)

Microsoft Certified: Azure AI Fundamentals — Microsoft (Jul 2021)

McKinsey Forward Program — McKinsey & Company (Apr 2024)

Circl Leader Coach Programme — Association of Coaching (Feb 2024)

Technical Skills

Programming Languages: C++, Java, Python, Embedded-C, MATLAB

DevOps Tools: Ansible, Docker, Kubernetes, Terraform, GitHub Actions, ArgoCD, Istio, Prometheus, Grafana

Cloud Platforms: AWS, GCP

Web Frameworks: Java Spring Boot, Angular

Databases: MySQL, DynamoDB

Other: Git/GitHub, Bash scripting, Linux administration (RHCSA), Windows Server (MCSA)